

Uva Wellassa University of Sri Lanka

Faculty of Applied Sciences

Department of Computer Science and Informatics

CST 262-2 Operating Systems concepts and Compiler Design

Tutorial 01

First-Come, First-Served (FCFS) & Convoy Effect

1. Consider a system running five processes. P1 is a heavily CPU-bound process, while P2,P3,P4, and P5 are short, interactive, I/O-bound processes.

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	40
P2	1	2
P3	1	1
P4	2	3
P5	3	2

- a) Draw the Gantt chart for this schedule.
- b) Calculate the Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), and Response Time (RT) for each process.
- c) Calculate the Average Waiting Time (AWT) and Average Turnaround Time (ATAT).
- d) Based on your results, explain the Convoy Effect observed here and discuss why FCFS is generally inefficient for modern time-sharing systems.

Shortest Job First (SJF) – Non-Preemptive vs. Preemptive (SRTF)

2.

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	10
P2	2	4
P3	3	2
P4	5	1
P5	6	4

- a) Case I (Non-Preemptive SJF): Draw the Gantt chart and compute the AWT and ATAT.
- b) Case II (Preemptive SJF / SRTF): Draw the Gantt chart and compute the AWT and ATAT.
- c) Compare the AWT of both cases. Justify mathematically which scheme optimizes performance for this process batch, and explain the main disadvantage long-running processes face under SRTF.

Priority Scheduling with Dynamic Aging

3. A system utilizes a Preemptive Priority scheduling algorithm.

Note: Lower integer value denotes higher priority (e.g., 1=highest priority, 5=lowest priority).

Aging is applied only to processes waiting in the ready queue, not to the currently executing process.

The priority of any process waiting in the ready queue is upgraded by 1 level (its integer value decreases by 1) every 3 ms of continuous waiting.

Process	Arrival Time (ms)	Burst Time (ms)	Initial Priority
P1	0	9	3
P2	1	4	4
P3	2	2	1
P4	6	5	5

- Draw the final Gantt chart for this preemptive schedule.
- Determine the Average Waiting Time (AWT) for the batch.

Round Robin (RR)

4. Five processes are scheduled using the Round Robin algorithm with a Time Quantum (q) = 4 ms.

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	10
P2	2	5
P3	4	3
P4	5	6
P5	8	2

- Show the step-by-step state of the Ready Queue over time as processes enter, execute, and get appended back to the tail.
- Draw the resulting Gantt chart.
- Calculate the Completion Time (CT), Waiting Time (WT), and Turnaround Time (TAT) for all processes.